

Endsem

Q1.

$$I = - \frac{dQ}{dt}$$

$$\Rightarrow \oint \vec{J} \cdot d\vec{S} = - \frac{d}{dt} \int \rho dV$$

↑ If this circle isn't present
-0.5

$$\Rightarrow \int (\vec{\nabla} \cdot \vec{J}) dV = - \int \left(\frac{\partial \rho}{\partial t} \right) dV$$

$$\Rightarrow \vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

It represents law of conservation of charge.

2

1

Q2.

(a) Characteristic impedance = $\sqrt{\frac{L}{C}}$

(20)

$$= \sqrt{\frac{227 \times 10^{-9}}{90.9 \times 10^{-12}}} \Omega$$

$$= 49.97 \Omega$$

$$(b) \text{ speed} = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{227 \times 10^{-9} \times 90.9 \times 10^{-12}}} \text{ m/s}$$

$$= 2.2 \times 10^8 \text{ m/s}$$

$$(c) \quad v = \frac{\omega}{\beta} = \frac{2\pi f}{\beta}$$

$$\Rightarrow \beta = \frac{2\pi f}{v} = \frac{2\pi \times 147 \times 10^8}{2.2 \times 10^8} \text{ rad/s}$$

$$= \frac{28\pi^2}{2.2} \text{ rad/s}$$

$$= 125.48 \text{ rad/s}$$

$$(d) \quad \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{30 - 49.97}{30 + 49.97}$$

$$= -0.25$$

$$(e) \text{ VSWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = \frac{1 + 0.25}{1 - 0.25}$$

$$= 1.67$$

Q4. (a) $n_1 = 1.5, n_2 = 1$

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2} = \frac{1.5 - 1}{1.5 + 1} = \frac{\frac{1}{2}}{\frac{5}{2}}$$

$$= \frac{1}{5}$$

$$= 0.2$$

(b) since $\Gamma > 0$, electric field does not
 flip sign. [No marks if part (a) is wrong. No marks if $\Gamma > 0$ isn't specified as reason]

$$(c) \quad T = \frac{2n_1}{n_1 + n_2} = \frac{2 \times \frac{3}{2}}{1 + \frac{3}{2}} = \frac{2 \times \frac{3}{2}}{\frac{5}{2}}$$

$$= \frac{6}{5}$$

$$= 1.2$$

$$(d) \quad SWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + \frac{1}{5}}{1 - \frac{1}{5}} = \frac{\frac{6}{5}}{\frac{4}{5}}$$

$$= \frac{3}{2} = 1.5$$

Q5. (a) (iv) $\rightarrow H_{t1} = H_{t2}$

(b) (iii) $\rightarrow$ Solenoidal

(c) (iii) $\int (\vec{\nabla} \cdot \vec{E}) dV = \oint \vec{E} \cdot d\vec{S}$

(d) (i) and (iii)

Q6(a) Inside $E=0$. So, Energy density = 0 1

Outside: $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$

$\therefore$ Energy density = $\frac{1}{2}\epsilon_0 \cdot E^2$.

$$= \frac{1}{2} \epsilon_0 \times \left(\frac{q}{4\pi\epsilon_0} \right)^2 \frac{1}{r^4}$$

$$= \frac{\epsilon_0 q^2}{32\pi^2} \left(\frac{1}{r^4} \right)$$

1

(b)

Total energy

$$= \frac{q^2 \epsilon_0}{32\pi^2} \int_{r=R}^{\infty} \int_0^{\pi} \int_0^{2\pi} \frac{1}{r^4} r^2 \sin\theta \, d\theta \, d\phi$$

A word
1 mark if the
integral
is written
correctly.

$$= \frac{q^2 \epsilon_0}{32\pi^2} \int_R^{\infty} \frac{1}{r^2} dr \int_0^{\pi} \sin\theta \, d\theta \int_0^{2\pi} d\phi$$

$$= \frac{q^2 \epsilon_0}{32\pi^2} \times 4\pi \left[-\frac{1}{r} \right]_R^{\infty}$$

$$= \frac{q^2 \epsilon_0}{8\pi} \left[\frac{1}{r} \right]_{\infty}^R = \frac{q^2 \epsilon_0}{8\pi R}$$

Q3.

$$f = 3 \text{ GHz} = 3 \times 10^9 \text{ Hz}$$

$$\omega = 2\pi f = 6\pi \times 10^9 \text{ rad/s}$$

Again this is crucial, if $3 \times 10^9 \text{ Hz}$ is used as ω , no marks in part (a). But 50% marks will be given in part (b) and (c), provided they are correct.

$$(a) \epsilon_{\text{eff}} = \epsilon \left(1 - i \frac{\sigma}{\omega \epsilon} \right)$$

$$= \epsilon - i \frac{\sigma}{\omega} = 24 \times 10^{-12} - i \frac{7.2 \times 10^4}{6\pi \times 10^9}$$

$$= (24 \times 10^{-12} - i 3.82 \times 10^{-6})$$

$$(b) \frac{E_x}{H_y} = \frac{E_z}{H} = \sqrt{\frac{\mu_0}{\epsilon_{\text{eff}}}}$$

$$= \sqrt{\frac{4\pi \times 10^{-7}}{24 \times 10^{-12} - i 3.82 \times 10^{-6}}} \text{ ohm}$$

$$= \sqrt{4\pi \times 10^{-7} (1.645 + i 2.618 \times 10^5)} \text{ } \Omega$$

$$\approx \sqrt{i 4\pi \times 10^{-2} \times 2.618} \text{ } \Omega$$

$$= \sqrt{0.329 i} \text{ } \Omega$$

$$= (0.4056 + i 0.4056) \Omega$$

2 marks till this point

*** Imp: here $\frac{E_x}{H} = \frac{E_z}{H_y}$

$$\neq \sqrt{\frac{\mu_0}{\epsilon_{\text{eff}}}}$$

So, simply calculating directly $\text{mod}(\sqrt{\frac{\mu_0}{\epsilon_{\text{eff}}}})$ won't fetch any marks.

$$(c) \quad k = k' - ik'' = \omega \sqrt{\mu_0 \epsilon_{eff}}$$

$$= 6\pi \times 10^9 \sqrt{4\pi \times 10^{-7} \times (2.4 \times 10^{12} - i 3.82 \times 10^{-6})}$$

Someone writes this correctly

2 marks

(But if the ω is incorrect and the calculation is left here, 0 marks)

$$\approx 6\pi \times 10^9 \sqrt{-i 4\pi \times 3.82 \times 10^{-13}}$$

$$= 6\pi \times 10^9 (1.549 - i 1.549) \times 10^{-6}$$

$$= 6\pi \times 10^3 (1.549 - i 1.549)$$

$$= 1.885 \times 10^4 (1.549 - i 1.549)$$

$$= (2.92 - i 2.92) \times 10^4$$

Q7.

$$V = A \frac{e^{-\lambda r_0}}{r_0}$$

Must be negative on both sides. Otherwise -1

$$\vec{F} = - \vec{\nabla} V$$

$$= - A \hat{r}_0 \frac{\partial}{\partial r_0} \left(\frac{e^{-\lambda r_0}}{r_0} \right)$$

$$= - A \hat{r}_0 \left(-\lambda \frac{e^{-\lambda r_0}}{r_0} - \frac{e^{-\lambda r_0}}{r_0^2} \right)$$

$$= A \hat{r}_0 \left(\frac{\lambda e^{-\lambda r_0}}{r_0} + \frac{e^{-\lambda r_0}}{r_0^2} \right)$$

2 marks (no step marking)

$$\rho = \vec{\nabla} \cdot \vec{E}$$

$$\epsilon_0 = \left[\vec{\nabla} \cdot \left(\frac{\lambda e^{-\lambda r}}{r^2} \hat{r} \right) + \vec{\nabla} \cdot \left(\frac{e^{-\lambda r}}{r^2} \hat{r} \right) \right] A \epsilon_0$$

$$= \left[\frac{\lambda}{r^2} \frac{\partial}{\partial r} \left(\frac{r^2 e^{-\lambda r}}{r} \right) + e^{-\lambda r} \vec{\nabla} \cdot \left(\frac{\hat{r}}{r^2} \right) + \frac{\hat{r}}{r^2} \cdot (\vec{\nabla} e^{-\lambda r}) \right] A \epsilon_0$$

← Award 0.5 if this is correct

$$= \left[\frac{\lambda}{r^2} (e^{-\lambda r} - \lambda r e^{-\lambda r}) + e^{-\lambda r} 4\pi \delta^3(r) + \frac{1}{r^2} (-\lambda) e^{-\lambda r} \right] A \epsilon_0$$

$$= \left[e^{-\lambda r} 4\pi \delta^3(r) - \frac{\lambda^2}{r} e^{-\lambda r} \right] A \epsilon_0$$